

BRADY LIN

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EDUCATION

Northwestern University
B.S. Electrical Engineering

Evanston, IL
Expected Jun 2028

SKILLS

Embedded development: C, C++, Python, Linux, RTOS, I2C, SPI, UART, interrupts

Hardware debugging: multimeter, serial monitor, oscilloscope, logic analyzer, JTAG

CAD and EDA: LTspice, MATLAB, KiCAD, EasyEDA, PCB layout, SolidWorks, Onshape, Fusion360

Fabrication: CNC, additive manufacturing, soldering (THT and SMD), laser/plasma cutting, waterjet, machining, most shop tools

Language: English (native), Mandarin (native), experienced in communicating with manufacturers in both languages

EXPERIENCES

Formlabs

Somerville, MA

Mechatronics Intern

Dec 2025 - March 2026

- Owned end-to-end development of an embedded camera + lighting system for a next-gen SLA 3D printer, including PCB design, mechanical integration, and embedded driver firmware in C
- Designed force-sensing fixtures to analyze deflection in resin-mixing arms, informing design decisions for coming printer prototypes
- Prototyped and iterated on a cam latch in Onshape with 3D printing, integrating Hall-effect sensors to reliably detect locking state

Northwestern University | Center for Robotics and Biosystems

Evanston, IL

Robotic Systems Engineer

March 2025 – Present

- Developed firmware on ESP32 platforms for closed-loop stepper motor control and I²C networking across 20+ nodes, enabling operation of a 1.5 m × 1.5 m CoreXY gantry for a 150-drone battery-swap array
- Improved system accuracy by reducing X-axis racking error from ~8° to less than 0.5° through PID tuning and kinematics analysis
- Designed 30+ parts in SolidWorks using DFM/DFA; fabricated via 3D printing, laser cutting, water jetting, and machining
- Validated embedded–mechanical integration by testing custom components across motion accuracy, load handling, and responsiveness

Creative Engineering & Robotic Arts Studio

Evanston, IL

Founder & Lead Engineer

Oct 2024 – Present

bradylin.com/projects/pulse/

- Led design and development of a six-armed robotic drummer integrating mechanical arms, animatronics, and LCD feedback, powered by a Linux computer accessed via SSH and a Teensy 4.1 running low-latency code
- Developed firmware in C++ to coordinate servos, solenoids, and steppers, synchronizing MIDI playback with zero effective delay through a real time pre-parser
- Optimized hardware–software integration by designing and validating 60+ mechanical, grounding, and thermal components in CAD
- Produced a live performance for 100+ attendees, overseeing cross-team collaboration to showcase the robot playing with a band.

PROJECTS

Expressive Violin-Style MIDI Instrument | Designer & Hardware Engineer

Nov 2024 – July 2025

bradylin.com/projects/stradex/

- Built a violin-inspired MIDI controller with bare metal C++ firmware on a custom PCB and Onshape-modeled enclosure
- Designed a compact PCB with 70+ components in KiCAD, integrating multiple I²C chips, sensors, and an RP2350 microcontroller
- Optimized firmware using DSP techniques and input hysteresis to ensure smooth and intuitive use.
- Validated reliability of 3D printed moving parts through 30+ rapid prototyping iterations and automated >10,000 actuation test cycles
- Released a project video that gained 16,000+ YouTube views, was featured in tech media such as Hackaday, and led to 20 unit sales

Low Level Computer-Vision Robotics Project | Mechatronics Engineer

March 2025 – May 2025

- Engineered a low-level computer vision line-following robot by writing Raspberry Pi Pico C-SDK firmware with PID control for dual DC motors, winning 1st place in the project-based graduate level class *Advanced Mechatronics, MECH_ENG 433*
- Optimized real-time image processing by implementing an SPI camera pipeline, tuning a PID controller, and building a centroid-based tracking algorithm, enabling fast lap times and stable path-following performance
- Debugged inter-module communication using logic analyzers and oscilloscopes, catching multiple algorithm-breaking bugs